Problem Statement

Cadmium Problems

Industrial activity is a source of some of the most toxic heavy metals, such as Cadmium (Cd^{2+}). It is not degraded, unlike organic pollutants. It is not broken down when it enters soil or water, and is bioaccumulated in crops and up the food chain to people. Cadmium is a long-term public health problem because it causes kidney damage, bone demineralization (known as itai-itai disease), and long-term cancer risks; it is an issue in industrialized and agricultural areas.

Cadmium Contamination in Taiwan is a National Problem.

The first cadmium rice-related incidents were discovered in the 1980s, leading Taiwan to undertake a long-term national survey of cadmium and other eight heavy metal (cadmium, lead, copper, zinc, nickel, arsenic, chromium and mercury) polluted farmland, which has been conducted by the Ministry of Environment (formerly EPA) and the Ministry of Agriculture.

The contamination is highly localized in the three counties/cities, which represent more than 98% of the total officially reported contaminated farmland:

Region	Share of Listed Farmland	Source of Contamination
Changhua County	~47%	Dense electroplating/metal processing industry interlaced with farmland; wastewater discharged into irrigation canals
Taoyuan City	~46%	Origin of the original cadmium rice incidents; chemical plants historically discharged high-cadmium wastewater into local streams and irrigation channels
Taichung City	~5%	Scattered discharge from metal surface treatment and unregistered factories

In Taiwan, the regulation limits acceptable concentration levels are set as follows:

- Monitoring standard: 2.5 mg/kg — indicates that a contamination issue is beginning to occur, and warrants more intensive water quality monitoring.
- The standard of 5.0 mg/kg is used to identify a "polluted control site" to trigger fallowing and ban on the production of food crops.

In a large scale investigation of a contaminated irrigation zone in Taoyuan's Luzhu district, cadmium contamination is found in 30.84% of 139 samples of brown rice, which were tested to determine cadmium levels above the applicable cadmium safety standard at the time (0.50

ppm). Much of this farmland has been remediated over the years, such as soil dilution, and "clean soil replacement" (removal and replacement of contaminated topsoil), but irrigation water in these areas is still being monitored on a long-term basis, and the original industrial wastewater sources are still a risk.

Why conventional remediation strategies don't work.

Traditional heavy-metal remediation technologies all have numerous limitations:

Chemical precipitation is effective but expensive, produces toxic sludge which has to be disposed of, and can add secondary contaminants (e.g., aluminum) to the environment.

There are ion exchange and physical adsorption techniques, which are effective, but require a high consumption of consumable items, thereby increasing the operating costs in the long run.

Conventional bioremediation with naturally-occurring microbes is more sustainable, but wild-type microbes are simply not efficient or targeted enough to satisfy industrial treatment requirements, and the majority are known to enter the cell to sequester the toxin, such as cadmium, in their interior, where it disrupts the normal metabolism of the cell and raises cell death.

This presents a compelling opportunity: finding a resource that is both low-cost and environmentally sustainable, has a high specificity for cadmium, and generates minimal secondary waste is not possible with any currently available solution.

The need for a Smarter, Biological solution.

There is a need for a system that can:

Selectively detect cadmium without being interfered with by chemically similar metals such as zinc or lead.

Effectively remove cadmium without allowing the contaminant to enter the cell, thus compromising cell functioning

Function at a low cost and on a large scale, without the use of expensive inducers, special equipment or toxic waste products

This is where our project comes in: an engineered E. coli system that detects and binds cadmium on the surface of the cell, intercepting the cadmium before it can enter the cell or further up the food chain.